

MetaFVG Trend-Filter Family: Regression & Lasso Formulations

Research notes — `metalib/research/`

Reference setup: HTF=4h, LTF=M15, post lookahead-bias fix

Notation

Let $c_1, \dots, c_n$ denote the HTF (4h) close series and, at bar i , let $y = (c_{i-w+1}, \dots, c_i)$ be the trailing window of width w (default $w = 20$), regressed against the bar-index regressor $x = (0, 1, \dots, w-1)$. All quantities below are computed causally: the value attributed to bar i uses only $c_{\leq i}$.

Meaning of the regressors. x is *not* calendar time — it is a purely local index that resets to 0 at the start of every window and counts up to $w-1$. Every rolling fit is therefore “how has price moved over the last w bars,” never an absolute-time trend. $d \in \{+1, -1\}$ throughout denotes the candidate trade direction already implied by the HTF FVG zone the current price sits in (from the signal pipeline, not from the regression itself) — every gate below only asks whether the regression *agrees* with that direction.

1 Regression Gate

A rolling univariate OLS of price on bar index within each trailing window:

$$c_t = \beta_0 + \beta_1 x_t + \varepsilon_t, \quad \hat{\beta}_1 = \frac{\sum_t (x_t - \bar{x})(y_t - \bar{y})}{\sum_t (x_t - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$
$$R^2 = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}}, \quad \text{SS}_{\text{res}} = \sum_t (y_t - \hat{y}_t)^2, \quad \text{SS}_{\text{tot}} = \sum_t (y_t - \bar{y})^2$$

Gate condition (a signal in direction $d \in \{+1, -1\}$ is accepted iff):

$$\text{sign}(\hat{\beta}_1) = d \quad \text{and} \quad R^2 \geq \tau_{R^2}, \quad \tau_{R^2} = 0.5 \text{ (default)}$$

Meaning of the regressors. $c_t = y_t$ is simply the price at bar t within the window — the thing being explained. x_t is that bar’s position in the window (see Notation): the regression asks “as x counts up from 0 to $w-1$, does price move consistently in one direction?” $\hat{\beta}_1$ (the slope) is the fitted *price change per bar* — its sign is the trend direction, its magnitude is irrelevant to the gate (only used to determine sign). $R^2 \in [0, 1]$ is the fraction of the window’s price variance explained by that straight line — a “cleanliness” score: R^2 near 1 means price moved almost monotonically, near 0 means it was choppy/range-bound even if the fitted slope happens to point the right way. $\tau_{R^2} = 0.5$ is the minimum cleanliness required before the gate trusts the direction call at all. This is the config that survived the HTF lookahead-bias fix with a real, if modest, edge (avg. individual Sharpe +0.010, equal-weighted 7-pair FX Majors portfolio Sharpe +0.143).

2 Theil-Sen Gate

Identical R^2 gate to §1, but the direction call uses the Theil–Sen (median-of-pairwise-slopes) estimator instead of the OLS slope — robust to a single outlier bar dominating the fit:

$$\hat{\beta}_1^{\text{TS}} = \text{median} \left\{ \frac{y_j - y_i}{j - i} : 0 \leq i < j < w \right\}$$
$$\text{Gate: } \text{sign}(\hat{\beta}_1^{\text{TS}}) = d \quad \text{and} \quad R^2 \geq \tau_{R^2}$$

Meaning of the regressors. For every pair of bars $i < j$ in the window, $(y_j - y_i)/(j - i)$ is the average price-change-per-bar *measured only between those two bars* — i.e. “if price only moved between these two points, what would the slope be.” Taking the median over all $\binom{w}{2}$ such pairs means one freak bar (a single large spike) can shift the result by at most one rank, unlike OLS where it can dominate the whole fit through the squared-error term. R^2 here is still the *OLS* goodness-of-fit from §1 — only the direction call changes, the cleanliness gate does not. Empirically tied Regression Gate almost exactly (avg. Sharpe +0.072 vs. +0.073 pre-refinement scale), i.e. outlier bars were not materially flipping the OLS direction call on this asset set.

3 Spearman Gate

Replaces both the OLS slope *and* R^2 with a single rank-correlation statistic, testing monotonicity rather than linearity (a nonlinear but consistently directional move is not penalized for curvature the way R^2 penalizes it):

$$\rho = 1 - \frac{6 \sum_t d_t^2}{w(w^2 - 1)}, \quad d_t = \text{rank}(y_t) - t$$

$$\text{Gate: } \text{sign}(\rho) = d \quad \text{and} \quad |\rho| \geq \tau_\rho$$

Meaning of the regressors. $\text{rank}(y_t)$ is y_t ’s rank (1st-smallest, 2nd-smallest, ...) among the w prices in the window; t itself is already the bar’s rank in time (since $x = 0, \dots, w - 1$ is already sorted). d_t is therefore the gap between “where does this bar rank by price” and “where does this bar rank by time” — if price rose monotonically, the two rankings are identical and every $d_t = 0$. $\rho \in [-1, 1]$ (Spearman’s rank correlation) summarizes that agreement in one number: +1 means price and time are perfectly rank-ordered together (monotonic uptrend, any shape), -1 means perfectly monotonic down, 0 means no consistent ordering. Unlike R^2 , ρ does not care whether the monotonic move was a straight line or a curve — only that it never reversed rank. Tested at $\tau_\rho \in \{0.5, 0.82, 0.85\}$; none beat Regression Gate on FX Majors post-fix (avg. Sharpe -0.011 to -0.254 depending on threshold and whether risk-sizing is layered on top).

4 Lasso Trend Gate (walk-forward)

A genuinely predictive, out-of-sample model: fit on a trailing window of `train_bars` (450) bars, refit every `test_bars` (190) bars, predicting the forward k -bar ($k = 5$) log return:

$$Y_t = \ln c_{t+k} - \ln c_t$$

Regressor vector at bar t (base feature set, before EMA augmentation):

$$X_t = [\text{slope}_t, R_t^2, \sigma_t^{\text{short}}, \sigma_t^{\text{long}}, \rho_t^{\text{vol}}, \text{slope}_t \rho_t^{\text{vol}}, R_t^2 \rho_t^{\text{vol}}, \text{slope}_t \sigma_t^{\text{short}}, 1]$$

where $\sigma_t^{\text{short}}, \sigma_t^{\text{long}}$ are realized volatility of log returns over 10- and 60-bar trailing windows, and $\rho_t^{\text{vol}} = \sigma_t^{\text{short}} / \sigma_t^{\text{long}}$. Fit by L_1 -penalized least squares (numba coordinate descent — this environment’s BLAS crashes on any matrix-matrix multiply, so no `sklearn/statsmodels` closed-form solve):

$$\hat{\beta} = \arg \min_{\beta} \frac{1}{2n} \|Y - X\beta\|_2^2 + \alpha \|\beta\|_1, \quad \alpha \text{ chosen by 5-fold CV over 60 candidate values}$$

Gate condition, with $\hat{Y}_t = X_t^\top \hat{\beta}$ (clipped to the training target range) and τ_t a *causal* rolling quantile of $|\hat{Y}|$ shifted one bar (never uses a prediction generated at or after t):

$$\text{sign}(\hat{Y}_t) = d \quad \text{and} \quad |\hat{Y}_t| \geq \tau_t$$

Meaning of the regressors. Y_t , the *target* being predicted, is “the log return you would realize if you held from now (t) to $k = 5$ HTF bars later” — this is the one genuinely forward-looking quantity in the whole document, and it is only ever used as a training label, computed from *fully elapsed* history at fit time (never a live, in-progress bar). Reading X_t component by component: slope_t, R_t^2 are exactly §1’s OLS outputs, but here they are *inputs* to a model rather than a hard pass/fail gate — the Lasso decides how much weight (if any) they deserve. σ_t^{short} (10-bar) and σ_t^{long} (60-bar) are realized volatility of log returns — “how much has price been jumping around lately,” measured over a quick and a slower window respectively. $\rho_t^{\text{vol}} = \sigma_t^{\text{short}} / \sigma_t^{\text{long}}$ is a *vol-regime* signal: > 1 means volatility is currently expanding relative to its own recent baseline, < 1 means it is contracting. The three products ($\text{slope}_t \rho_t^{\text{vol}}, R_t^2 \rho_t^{\text{vol}}, \text{slope}_t \sigma_t^{\text{short}}$) are curated interaction terms, e.g. “does a strong trend behave differently when volatility is expanding vs. contracting” — a small, hand-picked set rather than a full polynomial expansion, so the model stays interpretable by inspection of which coefficients the Lasso keeps nonzero. The trailing constant 1 is the intercept regressor. $\hat{\beta}$ is the fitted coefficient vector: the L_1 penalty shrinks the coefficient on any feature that doesn’t earn its keep exactly to zero, so $\hat{\beta}$ itself is a readable statement of “which of these nine things actually mattered this refit cycle.” α controls how harshly that shrinkage is applied (larger $\alpha \rightarrow$ fewer, smaller coefficients, chosen by cross-validation rather than fixed by hand). $\hat{Y}_t = X_t^\top \hat{\beta}$ is the model’s forecast of Y_t using only information available at t ; τ_t is a magnitude bar on that forecast — only trust predictions that are unusually large relative to the model’s own recent prediction history, not just any nonzero sign.

5 EMA-Augmented Lasso — Price-Based (superseded)

First attempt at adding EMAs as regressors: normalized price-vs-EMA distance and crossover, appended to X_t above, with EMA₉, EMA₂₁ computed on *price*:

$$\begin{aligned} \text{ema_fast_dist}_t &= \frac{c_t - \text{EMA}_9(c)_t}{c_t}, & \text{ema_slow_dist}_t &= \frac{c_t - \text{EMA}_{21}(c)_t}{c_t} \\ \text{ema_cross}_t &= \frac{\text{EMA}_9(c)_t - \text{EMA}_{21}(c)_t}{c_t}, & \text{plus interaction } &\text{ema_cross}_t \cdot \rho_t^{\text{vol}} \end{aligned}$$

Meaning of the regressors. EMA₉(c) and EMA₂₁(c) are exponentially-smoothed price — a “fast” running average that reacts within about 9 bars and a “slow” one that reacts over about 21 (the classic short/long CTA pairing; see §6 for the exact recursion). Dividing by c_t turns each raw price-minus-average into a *normalized* distance so it is comparable across instruments/price scales: $\text{ema_fast_dist}_t > 0$ means price sits above its own short-run average (recently strong), and similarly for the slow one against the longer-run average. ema_cross_t is the gap between the fast and slow average itself, again normalized by price — positive when the fast average has pulled above the slow one, the textbook “golden cross” direction signal. Underperformed Regression Gate at both tested thresholds (avg. Sharpe -0.308 loose, -0.257 tight; portfolio blend *worse* than the individual average in both cases, unlike Regression Gate where blending helps).

6 EMA-Augmented Lasso — Vol-Adjusted Returns (current)

Per your request: the EMA is taken of *vol-adjusted returns*, not price, so the smoothed series is comparable across volatility regimes:

$$\begin{aligned} r_t &= \ln c_t - \ln c_{t-1}, & r_t^{\text{vadj}} &= \frac{r_t}{\sigma_t^{\text{short}}} \\ \text{ema_fast}_t &= \text{EMA}_9(r^{\text{vadj}})_t, & \text{ema_slow}_t &= \text{EMA}_{21}(r^{\text{vadj}})_t \\ \text{ema_cross}_t^{\text{vadj}} &= \text{ema_fast}_t - \text{ema_slow}_t, & \text{plus interaction } &\text{ema_cross}_t^{\text{vadj}} \cdot \rho_t^{\text{vol}} \end{aligned}$$

where, for a series z , $\text{EMA}_p(z)_t = wz_t + (1 - w)\text{EMA}_p(z)_{t-1}$, $w = 2/(p + 1)$, seeded from the first non-NaN value and held steady across any NaN gap.

Meaning of the regressors. r_t is the plain bar-to-bar log return — “how much did price move this bar,” with no normalization. r_t^{vadj} divides that by σ_t^{short} (the same 10-bar realized vol from §4), turning it into “how many typical-recent-moves was this bar’s return,” roughly a rolling z-score of the return: a 0.5% move counts as large during a calm stretch and as unremarkable during a volatile one, whereas a raw-price EMA cannot distinguish the two. $\text{ema_fast}_t, \text{ema_slow}_t$ are no longer distances from anything — since the base series is already a (vol-adjusted) return, not a price level, the smoothed value itself *is* the signal: a running estimate of recent vol-adjusted momentum, fast (9-bar) and slow (21-bar). $\text{ema_cross}_t^{\text{vadj}}$ is their difference, i.e. momentum *acceleration* — positive when recent momentum is running hotter than its own longer-run average. In the smoothing recursion, $w = 2/(p + 1)$ is the EMA weight (not to be confused with the regression window width w used in §1–§3 — unfortunately the same letter, distinguished by context in the source). Recovers some ground versus the price-based version but still loses to Regression Gate (avg. Sharpe -0.239 loose, -0.230 tight; blend still worse than the individual average).

7 Shelved: LTF Significance-Gated OLS Bias

From the (rejected) “back to basics” redesign: same closed-form OLS as §1, but applied to the *LTF* (M15) close series, and gated on statistical significance of the slope rather than goodness-of-fit:

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum_t (x_t - \bar{x})^2}}, \quad \hat{\sigma}^2 = \frac{\text{SS}_{\text{res}}}{w - 2}, \quad t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)}$$

$$\text{Gate: } \text{sign}(\hat{\beta}_1) = d \quad \text{and} \quad |t| \geq \tau_t$$

Meaning of the regressors. $\hat{\sigma}^2$ is the fit’s residual variance — how noisy the window is around its own best-fit line, degrees-of-freedom-corrected. $\text{SE}(\hat{\beta}_1)$ turns that into an uncertainty estimate specifically for the slope: how much would $\hat{\beta}_1$ plausibly wobble if re-fit on similar noise. The t -statistic is the slope measured in units of its own uncertainty — “is this slope big compared to how unsure we are about it,” as opposed to R^2 in §1 which only asks “does a line fit well” without any notion of estimation uncertainty. τ_t is the minimum $|t|$ required to trust the direction call, analogous to τ_{R^2} in §1 but on a statistical-significance scale instead of a goodness-of-fit scale. Tested $\tau_t \in [0.5, 4.0]$ across all 7 FX Majors: average Sharpe was negative throughout and *worsened* as τ_t tightened (-0.286 at $\tau_t=1.5$ down to -0.370 at $\tau_t=3.5$) — the opposite of the single-symbol (EU-RUSD) read that motivated testing it, and GBPUSD flipped from this campaign’s best individual result (Sharpe 1.10 under §1) to one of this variant’s worst (-1.03). Shelved; code retained in `metaavg-basics-variant.py` for reference.

Summary

Config	Avg. individual Sharpe	Blended portfolio Sharpe
Regression Gate	+0.010	+0.143
Theil-Sen Gate	+0.072*	—
Spearman Gate (q=0.5)	−0.011*	—
Lasso Trend Gate (base features)	−0.16 to −0.42*	—
EMA-Augmented Lasso (price)	−0.308 / −0.257	−0.688 / −0.712
EMA-Augmented Lasso (vol-adj.)	−0.239 / −0.230	−0.488 / −0.640

* Full-universe (47-symbol) scale, not the FX-Majors-only re-verification used for the rows below it; included for context, not a like-for-like comparison.